

1. Infinite Monkey Theory 猴子打字理论 (多墨 69) (萤火虫 11) (猩际 62)

This lecture talks about the infinite monkey theorem. This theorem states that a monkey hitting keys at random on a typewriter keyboard for an infinite amount of time will almost surely type a given text. Such as the works of William Shakespeare. Almost surely is a mathematical term with a precise meaning, and the monkey is not an actual monkey but a metaphor for a device. So it's mathematically possible but incredibly unlikely to happen.

本讲座讨论了无限猴子定理。该定理指出，一只猴子在无限长的时间内随机敲击打字机键盘上的按键几乎肯定会打出给定的文本。例如威廉·莎士比亚的作品。几乎肯定是一个具有精确含义的数学术语，猴子不是真正的猴子，而是一种设备的隐喻。因此，这在数学上是可能的，但发生的可能性极小。

2. Haussmann's Renovation of Paris 奥斯曼对巴黎的改造 (萤火虫 7) (猩际 128)

This lecture mainly talks about the renovation of Paris and Napoleon. The renovation was a vast public program introduced by Napoleon the third. Napoleon the third told Haussmann to bring air and light to the center to make the city safer and more beautiful. The renovation removed the unhealthy neighborhood and it built roads, parks, and squares, planted more trees and constructed new infrastructures. Finally, the speaker mentions that the reason for doing this is that the old Paris had many serious problems such as overcrowding, disease, and crime.

本次讲座主要讲述巴黎的改造和拿破仑。改造是拿破仑三世发起的一项庞大的公共计划。拿破仑三世告诉奥斯曼，要把空气和阳光带到市中心，让城市更安全、更美丽。改造移除了不健康的街区，修建了道路、公园和广场，种植了更多的树木，建造了新的基础设施。最后，演讲者提到这样做的原因是旧巴黎存在许多严重的问题，例如过度拥挤、疾病和犯罪。

3. Galaxy Darkness 银河系黑暗之谜 (多墨 72) (猩际 139)

Space appears dark because the universe is not infinite, and there aren't enough stars to light up every part of it. If the universe were infinite, every line of sight would end at a star, creating a bright night sky, known as Olbers' Paradox. Early astronomers thought dust absorbed starlight, but this would cause the dust to glow, adding to the light. Now, scientists understand that our finite universe simply lacks the star density to illuminate the entire night sky. Other factors, such as the expansion of the universe, further contribute to the darkness we observe in space.

太空之所以显得黑暗，是因为宇宙不是无限的，也没有足够的恒星照亮宇宙的每个部分。如果宇宙是无限的，那么每条视线都会以恒星为终点，形成明亮的夜空，这就是所谓的奥尔伯斯悖论。早期天文学家认为尘埃吸收了星光，但这会导致尘埃发光，从而增加光亮。现在，科学家们明白，我们有限的宇宙根本没有足够的恒星密度来照亮整个夜空。其他因素，例如宇宙膨胀，进一步加剧了我们在太空中观察到的黑暗。

4. Welsh Speaker 威尔士语使用者 (多墨 74)

Welsh is a Celtic language spoken by approximately 740,000 people in Wales and several hundred in Patagonia, Argentina. There are also Welsh speakers in England, Scotland, the USA, Canada, Australia, and New Zealand. At the start of the 20th century, half of Wales's population spoke Welsh, but by the end of the century, only about 21% did. According to the 2001 census, 582,368 people in Wales could speak Welsh, with 28% having some knowledge of it. S4C reports around 750,000 Welsh speakers in Wales, with an additional 1.5 million who can understand the language.

威尔士语是一种凯尔特语，威尔士约有 74 万人使用这种语言，阿根廷巴塔哥尼亚也有数百人使用这种语言。英格兰、苏格兰、美国、加拿大、澳大利亚和新西兰也有人使用威尔士语。20 世纪初，威尔士一半的人口使用威尔士语，但到 20 世纪末，只有约 21% 的人使用这种语言。根据 2001 年的人口普查，威尔士有 582,368 人会讲威尔士语，其中 28% 的人对这种语言有所了解。S4C 报告称，威尔士约有 75 万威尔士语使用者，另外还有 150 万人能听懂这种语言。

5. Bomb Calorimeter 燃烧计测量原理 (多墨 75) (萤火虫 5) (猩际 72)

This lecture is mainly about bomb calorimeter. The equipment can help to measure how much energy contained in food. There will be a chemical combustion between the food and oxygen, and the water chamber absorbs the heat when food samples are burned. The thermometer is used to measure the increased temperature of the water, and then, the energy of the food can be calculated. But the result is

different from the energy human absorb from food because our digestion system cannot absorb all energy from food.

本次讲座主要讲弹式热量计，该设备可以帮助测量食物中含有多少能量。食物和氧气之间会发生化学燃烧，食物样品燃烧时水室会吸收热量。温度计用于测量水温的升高，然后可以计算出食物的能量。但结果与人类从食物中吸收的能量不同，因为我们的消化系统无法吸收食物中的所有能量。

6. Western Countries Low Childbirth Rate 西方国家出生率 (猩际 130)

The western countries women are becoming more and more reluctant to give birth to babies. The birth rates increased during the 20's century but it remains to decrease in the last two decades. In 2000, the birth rate remained at around 1%, and there are even some negative birth rates in other countries. The birth rate dropped to its lowest point that has never been seen in the history. It also has impacts on the male in the society especially young man and might have some connection with the unemployment rate as well.

西方国家的女性越来越不愿意生孩子，20 世纪出生率有所上升，但近 20 年来出生率持续下降，2000 年出生率维持在 1% 左右，其他国家甚至出现了负出生率，出生率降到了历史最低点，对社会男性尤其是年轻男性也有影响，可能和失业率也有一定关系。

7. Australian Export Business 澳洲与多国出口业务 (多墨 78) (萤火虫 1) (猩际 216)

This lecture talks about Australia's export. In the past, Australia was concerned about its geographical location, which may result in Australia being isolated from other continents. Nevertheless, with the rise of Asian countries, especially China, Australia has become a great export country with a perfect location. Currently, Japan is the largest exporting country to Australia, but China may become the largest one in the future. Australia should take the advantage of China's rise to develop its export.

本讲座讲的是澳大利亚的出口。过去，澳大利亚担心地理位置，可能导致澳大利亚与其他大洲隔绝。然而，随着亚洲国家，尤其是中国的崛起，澳大利亚已经成为一个地理位置优越的出口大国。目前，日本是澳大利亚最大的出口国，但未来中国可能会成为最大的出口国。澳大利亚应该利用中国的崛起来发展出口。

8. Frog's Variation 青蛙变异影响 (多墨 82) (萤火虫 12) (猩际 343)

This lecture talks about the frog's variation. The graph shows three types of frogs in a different region, indicating their life habits, and their influence on human. Some kinds of frogs living in fresh water, some kinds of them live on the dry land, underground and in the trees. The number of malformations of frog is continuously increasing. Some of them have more limbs, while others have less. This lecture also explains the reason for the gene mutation. People are worrying that these malformations would affect the quality of water and the health of human beings.

本讲座讲的是青蛙的变异。图表展示了不同地区的三种青蛙，表明它们的生活习性，以及对人类的影响。有些青蛙生活在淡水中，有些则生活在陆地、地下和树上。青蛙的畸形数量不断增加。有些青蛙的四肢较多，而有些则较少。本讲座还解释了基因突变的原因。人们担心这些畸形会影响水质和人类的健康。

9. Language Extinction 语言灭绝 (猩际 122)

This lecture talks about minor language extinction. A language dies when the last person who speaks it dies. Many minor languages are disappearing at a rate that was never seen before in history. There is an increasing trend of minor language extinction recently. The main reasons for the minor language extinction are globalization and urbanization. People who speak small language choose to move to urban areas, which are not suitable for small language to survive. A great deal can be done to preserve endangered language. But without money, endangered languages don't have a positive future.

本次讲座探讨了小语种的灭绝。当最后一个使用该语言的人去世时，该语言就消亡了。许多小语种正在以历史上前所未有的速度消失。最近，小语种的灭绝趋势越来越明显。小语种灭绝的主要原因是全球化和城市化。说小语种的人选择搬到城市地区，而城市地区并不适合小语种生存。可以做很多事情来保护濒危语言。但是没有资金，濒危语言的未来就很渺茫。

10. Human's Creativity 人类创造性 (多墨 98)

Creativity relies on three key factors: people, process, and product, with process being the most crucial. First, education fosters a creative mindset in people. Second, an effective process engages individuals in innovation. Third, identifying the right problem is essential. Creativity supports human survival and prosperity. A creative product reflects the creator's ingenuity and should be surprising, original, beautiful, and useful. To cultivate creativity, people need talent, the right mindset, and the ability to view ideas from fresh perspectives. Innovation expands conceptual spaces through imagination and insight, enabling unique and meaningful creations.

创造力依赖于三个关键因素：人、流程和产品，其中流程最为关键。首先，教育培养人们的创造性思维。其次，有效的流程让个人参与创新。第三，找到正确的问题至关重要。创造力支持人类的生存和繁荣。创意产品反映了创造者的独创性，应该令人惊讶、新颖、美观且实用。要培养创造力，人们需要天赋、正确的思维方式以及从新角度看待想法的能力。创新通过想象力和洞察力扩展概念空间，从而实现独特而有意义的创作。

11. Performance of Maths and English 男女学习差异 (多墨 103)

The two charts show differing performances between boys and girls in Math and English. In English, girls consistently score about 10 points higher than boys over six years. However, in Math, there is no significant gender difference. To explain this, researchers examine biological, cognitive, and social factors. These elements interact in early childhood and are reinforced in school, creating distinct gender patterns in behaviour and skills, affecting performance. Evidence suggests that cognitive development is closely tied to social factors, such as learned gender roles, which are reinforced by teacher responses, shaping these performance patterns.

两张图表显示了男孩和女孩在数学和英语方面的不同表现。在英语方面，女孩在六年内始终比男孩高出约 10 分。然而，在数学方面，性别差异并不明显。为了解释这一现象，研究人员研究了生物、认知和社会因素。这些因素在幼儿时期相互作用，并在学校得到强化，在行为和技能方面形成了不同的性别模式，从而影响了表现。有证据表明，认知发展与社会因素密切相关，例如习得的性别角色，这些角色通过教师的反应得到强化，从而塑造了这些表现模式。

12. Underwater Detector 水下探测器 (多墨 105)

This lecture talks about underwater Antares detectors, which is a camera installed under the water for fish monitoring. It is a remote technology system; some researchers can monitor fishes on their computer in university. The camera can detect the surrounding environment, water temperature change, and fish. This help finds out what they eat or not. They can stop and makes it possible for people to change their feeding strategy quickly and can stop feeding then if necessary in some situation to save time and labor.

本讲座将介绍水下 Antares 探测器，这是一种安装在水下用于监测鱼类的摄像头。这是一个远程技术系统；一些研究人员可以在大学的计算机上监测鱼类。摄像头可以检测周围环境、水温变化和鱼类。这有助于了解它们吃什么或不吃什么。它们可以停止进食，使人们能够快速改变喂食策略，并在某些情况下必要时停止喂食，以节省时间和劳力。

13. Brain Development 大脑发育三阶段 (多墨 109) (萤火虫 2) (猩际 66)

This lecture talks about three stages of brain development, which are the primitive brain, limbic brain, and neocortex. Different sections of the brain have different function and finished development at different times. For sensory pathways, including vision and hearing, develop from embryo time, peak at 3 months old, stop at more than 4 years old; for language, develop from embryo time, peak around 9 months old, and stop more than 4-year-old; for higher cognitive function, develop from embryo time, peak at 1 year old and develop for a longer period. Moreover, higher level functions such as logical thinking, based on a lower level of development.

本讲座讲的是大脑发育的三个阶段，即原始脑、边缘脑和大脑皮层。大脑的不同部分有不同的功能，发育完成的时间也不同。对于感觉通路，包括视觉和听觉，从胚胎时期开始发育，3 个月时达到顶峰，4 岁以上停止；对于语言功能，从胚胎时期开始发育，9 个月左右达到顶峰，4 岁以上停止；对于高级认知功能，从胚胎时期开始发育，1 岁时达到顶峰，发育时间较长。而逻辑思维等高级功能，则基于较低的开发水平。

14. Marshmallow Test 棉花糖实验 (多墨 112) (萤火虫 77) (猩际 123)

This lecture talks about the marshmallow test, which is a kind of resisting temptation. The kids are told that they can get two marshmallows if they can wait for 15 minutes without eating the one marshmallow. The result shows that the longer time child resists the treat, the higher competitive he or she will be later in life. Scientists found that the ability to resist temptation is not strictly innate, it is highly influenced by the environment. If the previous promise was denied, the ability to resist temptation got hampered later.

本讲座讲的是棉花糖测试，这是一种抵制诱惑的测试。孩子们被告知，如果他们能等上 15 分钟而不吃掉一颗棉花糖，他们就可以得到两颗棉花糖。结果表明，孩子抵制诱惑的时间越长，他或她以后的竞争力就越强。科学家发现，抵制诱惑的能力并不是天生的，它在很大程度上受到环境的影响。如果先前的承诺被拒绝，那么抵制诱惑的能力以后就会受到阻碍。

15. Civil Society Groups 公民社会团体 (多墨 120)

There is no decline in involvement of organizations which seek to share wealth and opportunities and protect human right. Civil society groups have grown 40 folds since last century. The non-profit sector is worth one trillion dollars, and there are 700,000 such organizations in Australia alone. The best predictor of governmental success was the strength and density of a region's civic associations. The UN recognizes 37 thousand organizations across the globe. The speaker emphasizes the importance of NGO, which is called “the third sector”.

致力于分享财富和机会、保护人权的组织的参与度没有下降。自上个世纪以来，民间团体的数量增长了 40 倍。非营利部门价值一万亿美元，仅在澳大利亚就有 70 万个这样的组织。预测政府成功的最佳指标是地区民间协会的实力和密度。联合国承认全球有 3.7 万个组织。演讲者强调非政府组织的重要性，它被称为“第三部门”。

16. Climate Change 气候变化 (多墨 1321) (狸际 141)

Climate change negatively impacts agriculture, making some lands unsuitable for crops and threatening food security, especially in Africa. Millions may face hunger as production declines, with food supplies predicted to decrease by 10-17% annually, a trend expected to continue through 2070. Developing countries, particularly in tropical areas, struggle with these challenges due to limited resources. Extreme weather, like floods and hurricanes, further worsens food production, impacting the global economy and hitting African nations the hardest. This crisis highlights the urgent need for solutions to address climate-driven food insecurity.

气候变化对农业产生负面影响，导致一些土地不适合种植农作物，并威胁到粮食安全，尤其是在非洲。随着产量下降，数百万人可能面临饥饿，预计粮食供应每年将减少 10-17%，这一趋势预计将持续到 2070 年。发展中国家，特别是热带地区的发展中国家，由于资源有限，正在努力应对这些挑战。洪水和飓风等极端天气进一步恶化了粮食生产，影响了全球经济，对非洲国家的打击最大。这场危机凸显了解决气候导致的粮食不安全问题的迫切需要。

17. Narratives/ Visual Description 叙述故事/视觉描述 (狸际 199)

The comics I show you with lots of people chatting around in a room is a form of description. We use different kinds of methods to describe a situation. Sometimes we have to use visual description, particularly when we do not witness the scene. I was born during the second world war. When I ask my mother about the war, I always ask her you have mentioned this or that when you talked to me. When asked her about the shelter, I asked her what the shelter looks like and when did you go to the shelter. From her response, I could get more visual evidence as I can to write my book.

我向你展示的漫画是许多人在房间里聊天，这是一种描述形式。我们使用不同的方法来描述一种情况。有时我们必须使用视觉描述，特别是当我们没有亲眼目睹现场时。我出生于二战期间。当我问我母亲关于战争的事情时，我总是问她你在我谈话时提到过这个或那个。当问到她关于避难所的事情时，我问她避难所是什么样子的，你什么时候去避难所的。从她的回答中，我可以得到更多的视觉证据来写我的书。

18. Australian Big Cities 澳洲城市 (多墨 1323)

This lecture talks about big cities in Australia. Australia is the most urbanized countries in the world. Australia location is important for the world's export. Its international trade is also important since Australia has a broad territory. The Australian people are mainly living in five cities: Sydney,

Melbourne, Perth, Adelaide, and Brisbane. The most special one in Perth, which is one of the most isolated cities in the world. And there are two leading companies such as Telstra and Qantas which are based in Perth.

本讲座讲的是澳大利亚的大城市。澳大利亚是世界上城市化程度最高的国家。澳大利亚的地理位置对世界出口至关重要。由于澳大利亚领土广阔，其国际贸易也很重要。澳大利亚人主要居住在五个城市：悉尼，墨尔本，珀斯，阿德莱德和布里斯班。最特殊的是珀斯，它是世界上最孤立的城市之一。而两家领先的公司 Telstra 和 Qantas 的总部都设在珀斯。

19. Human Behaviour 人类行为/内外因素 (多墨 1324) (萤火虫 104) (猩际 173)

The lecture talks about two determinants of human's behaviour, internal and external factors. The personal factors are internal, and the environmental factors are external. The internal factor is the personal factor, including people's belief and their individual thinking. The external factor is the environmental factor, including temperature, air pressure and how other people think about them. Human's behaviours are determined by themselves and the environment.

讲座讲的是决定人的行为的两个因素，即内在因素和外在因素。个人因素是内在的，环境因素是外在的。内在因素是个人因素，包括人们的信念和个人思维。外在因素是环境因素，包括温度、气压和其他人对他们的看法。人的行为是由自身和环境决定的。

20. Pavlov Experiment 巴普洛夫实验 V1 (萤火虫 8)

Pavlov's experiment led psychologists to new insights about animal behaviour and human nature, particularly in learning. Although initially focused on digestion, Pavlov, a Nobel Prize-winning Russian scientist, noticed an unexpected reaction in dogs. When food was presented, they salivated as expected. However, over time, the dogs began salivating at the sight of food, the food dish, or even the footsteps of Pavlov or his assistant. Curious about this response, Pavlov tested various stimuli like metronomes, lights, and bells. He found that any stimulus could trigger salivation if it consistently signalled that food was coming.

巴甫洛夫的实验让心理学家对动物行为和人类天性有了新的认识，特别是在学习方面。尽管最初关注的是消化，但这位获得诺贝尔奖的俄罗斯科学家巴甫洛夫注意到狗有一种意想不到的反应。当食物出现时，它们会像预期的那样流口水。然而，随着时间的推移，狗一看到食物、食物盘，甚至巴甫洛夫或他助手的脚步声，就开始流口水。巴甫洛夫对这种反应很好奇，于是测试了各种刺激物，如节拍器、灯光和铃铛。他发现，如果任何刺激物持续发出食物即将到来的信号，都可能引发唾液分泌。

21. Overweight Problem 超重问题 (萤火虫 24)

The lecture talks about the overweight problem. There are 20% of children today have the overweight problem, which brings the heart disease are more and more common in children, the smallest is 5 years old. The situation makes the heart attack and other health problems become earlier and earlier. This issue needs to be solved because the overweight problems will be in more serious situations such as Type 2 Diabetes, kidney failure and strokes.

讲座讲到超重问题。现今有 20% 的儿童有超重问题，这导致儿童患心脏病的现象越来越普遍，最小的只有 5 岁。这种情况使得心脏病发作和其他健康问题越来越早出现。这个问题需要解决，因为超重问题会导致更严重的后果，如 2 型糖尿病、肾衰竭和中风。

22. Black Hole 黑洞 (多墨 1328)

The simulation depicts what you might observe while orbiting a black hole. The black hole's gravity distorts and spins the light and position of stars. The constellation Orion, for instance, appears to approach the event horizon, where it separates and spins, only to reappear normally on the other side. Users can explore scenarios such as slowing descent with rocket engines or experiencing free fall. In the centre, the entire universe's light is concentrated into a bright ring, showcasing the intense gravitational effects of the black hole.

模拟描绘了绕黑洞运转时可能观察到的情况。黑洞的引力会扭曲和旋转恒星的光线和位置。例如，猎户座似乎正在接近事件视界，在那里它分离并旋转，最后在另一边正常地重新出现。用户可以探索火箭发动机减速下降或体验自由落体等场景。在中心，整个宇宙的光都集中成一个明亮的环，展示了黑洞强烈的引力效应。

23. Loggerhead Sea Turtles 红海龟 (多墨 1330) (萤火虫 40)

The lecture talks about loggerhead turtle, one of the largest turtles in the world, and almost extinct in the USA. They have big heads and short necks. Scientists put a tracker on a turtle's shell, and use satellites to track and locate the migration route of the turtle. The turtles reach different localities in a different time, the rough route is from South to North. The migration takes three months. Experts are now worried that rising temperatures could cause havoc for turtle populations by hatching too many girls. Scientists say it's not going to happen overnight.

讲座中谈到了红海龟，它是世界上最大的海龟之一，在美国几乎已经灭绝。它们的头很大，脖子很短。科学家在海龟壳上安装了一个追踪器，并使用卫星跟踪和定位海龟的迁徙路线。海龟到达不同地方的时间不同，大致路线是从南到北。迁徙需要三个月。专家们现在担心，气温升高可能会孵化出太多雌性海龟，从而对海龟种群造成破坏。科学家说这不会在一夜之间发生。

24. Engineers and Medical Machines 生物工程师 (多墨 1339) (猩际 242)

The biotech engineering has undergone three phases. In the first phase, people designed and made some simple machines, which aimed to help people understand ourselves. And first group of people that realized we could learn from ourselves were engineers. And then, based on the studies, people began to make complicated devices, to cure wounds, like the practice of suture, and to study physiology. This is the second phase. Later, since we went into the third phase, more complicated machines have been developed, like ECG (electrocardiography) and X-ray picture. These devices benefit humans greatly in solving health problems, such as diseases.

生物工程经历了三个阶段。在第一阶段，人们设计并制造了一些简单的机器，目的是帮助人们了解自己。第一批意识到我们可以向自己学习的人是工程师。然后，基于这些研究，人们开始制造复杂的设备，治疗伤口，比如缝合，并研究生理学。这是第二阶段。后来，自从我们进入第三阶段以来，人们开发出了更复杂的机器，比如 ECG（心电图）和 X 光片。这些设备在解决疾病等健康问题方面对人类大有裨益。

25. The Randomness of Flipping Coin 抛硬币随机性 (多墨 1345) (萤火虫 119) (猩际 185)

Flipping a coin may not be as fair as it seems. About a decade ago, statistician Persi Diaconis, with Harvard engineers, built a mechanical coin flipper to test the randomness of coin tosses. He found that if a coin is flipped consistently, it lands the same way every time, suggesting that human error introduces randomness. Variations in height, speed, and angle affect outcomes. Using high-speed cameras and equations, Diaconis also discovered a bias: if a coin starts heads-up, it's more likely to land heads-up, showing that even "random" coin flips have a slight bias.

抛硬币可能并不像看上去那么公平。大约十年前，统计学家 Persi Diaconis 与哈佛大学的工程师们一起制造了一个机械抛硬币器，用于测试抛硬币的随机性。他发现，如果一枚硬币被持续抛掷，每次落地时它都会以同样的方式落地，这表明人为错误带来了随机性。高度、速度和角度的变化会影响结果。使用高速摄像机和方程式，Diaconis 还发现了一个偏差：如果一枚硬币一开始是正面朝上的，那么它落地时正面朝上的可能性就更大，这表明即使是“随机”抛硬币也存在轻微的偏差。

26. Morton Prince 莫顿普林斯 / 多重人格 (多墨 1351) (萤火虫 23) (猩际 240)

Morton Prince was an American physician and psychologist; his book Dissociation of a Personality was the best-seller at that time. It tells a story of Miss Christine Beauchamp, who was suffering from MPD (Multiple Personality Disorder). Miss Beauchamp has several personalities, namely B1, B2 and B3. There was hidden memory in these 3 personalities. Miss Beauchamp was B2. B2 knows about B1, B3 knows both B1 B2, but B1 knows nothing about B2 or B3. The strongest personality account for most of the time and it will take over the others and become the main personality at the end. This case and theory give great help to crime investigation.

莫顿·普林斯是美国的一位医生和心理学家，他所著的《人格分裂》是当时最畅销的书。书中讲述了一位患有多重人格障碍的克里斯汀·波尚小姐的故事。波尚小姐有多重人格，分别是 B1、B2 和 B3，这三种人格中都有隐藏的记忆。波尚小姐是 B2，B2 知道 B1，B3 知道 B1 和 B2，但 B1 对 B2 和 B3 一无所知。最强的人格占据了大部分的时间，最终会取代其他人格成为主要人格。这个案例和理论对犯罪侦查有很大的帮助。

27. Gene and DNA 基因及遗传物质 (多墨 1630) (萤火虫 34)

In this lecture, the speaker talked about the gene and DNA. At the beginning of the lecture, the speaker mentioned that genes decide the protein in the cells. Genes produce and make proteins. The

process from genes to protein is complex. There are more than two million proteins in a cell, but they still not result in the functions of different cells. They are the fundamental building blocks of all organisms. Genes are developed, the genetic difference in cognitive ability between present and ancestor, but the change is little. DNA, shown in the picture, is like a chain.

在这次讲座中，主讲人谈到了基因和 DNA。讲座一开始，主讲人提到基因决定细胞中的蛋白质。基因产生并制造蛋白质。从基因到蛋白质的过程非常复杂。一个细胞中有超过两百万个蛋白质，但它们仍然不影响不同细胞的功能。它们是所有生物体的基本组成部分。基因是发达的，现在和祖先之间的认知能力存在遗传差异，但变化很小。图中的 DNA 就像一条链。

28. Dogs and Sound 狗与声音 (萤火虫 74) (猩际 56)

The lecture introduces an experiment showing that dogs use different barks and growls to communicate different things. Researchers recorded an alone bark, a stranger growl where a dog was growling at the approach of a stranger, and a food growl, where a dog was growling to protect its food. The researchers played these different growls to a dog who was approaching a plate of juicy food. The dogs were more hesitant to approach if they heard the food growl rather than the stranger growl or the alone barks. The experiment shows that dogs could clearly distinguish between different types of growls and react to them accordingly.

讲座介绍了一项实验，该实验表明，狗会使用不同的吠叫和咆哮来传达不同的信息。研究人员记录了单独的吠叫、陌生人咆哮（狗在陌生人靠近时发出的咆哮声）和食物咆哮（狗为了保护食物而发出的咆哮声）。研究人员向一只正在靠近一盘多汁食物的狗播放了这些不同的咆哮声。如果狗听到的是食物咆哮声，而不是陌生人的咆哮声或单独的吠叫声，它们会更犹豫不决。实验表明，狗可以清楚地区分不同类型的咆哮声并做出相应的反应。

29. Housing Price 澳大利亚房价 (多墨 1639)

Young Australians face significant challenges in buying homes due to low housing affordability, altering the typical housing cycle. Traditionally, Australians moved out in their late teens or 20s, rented while saving, then bought a home, paid off their mortgage by middle age, and enjoyed security in retirement. Now, young people often live with parents longer, rent for extended periods, and carry mortgages into retirement. In 2006, 65,000 retiree households still had mortgages. Affordable rental options are also scarce, with low vacancies and high turnover as landlords maximize profits, making long-term rentals uncommon compared to other countries.

由于住房负担能力低，澳大利亚年轻人在买房方面面临巨大挑战，改变了典型的住房周期。传统上，澳大利亚人在十几岁或二十几岁时搬出去，一边租房一边存钱，然后买房，到中年还清抵押贷款，退休后享受保障。现在，年轻人往往与父母同住更长时间，长期租房，退休后还要背负抵押贷款。2006 年，65,000 个退休家庭仍有抵押贷款。经济实惠的租赁选择也很少，空置率低，周转率高，因为房东追求利润最大化，与其他国家相比，长期租赁并不常见。

30. Oracle 甲骨文 (萤火虫 154)

Oracle cloud system, data are stored in a machine called database, all people can access the materials from this system. Oracle is the largest database company in the world. It has many services, such as email, voicemail, and database. It helps businesspeople travel around the world and do business anywhere. Mobile phone let people communicate on the way. In conclusion, technology provides people with a lot of conveniences.

Oracle 云系统，数据存储称为数据库的机器中，所有人都可以从该系统访问资料。Oracle 是世界上最大的数据库公司。它提供许多服务，例如电子邮件、语音邮件和数据库。它帮助商人环游世界并在任何地方做生意。手机让人们在途中交流。总之，技术为人们提供了许多便利。

31. Mars and Earth 火星和地球 (多墨 1644) (萤火虫 45)

This lecture compares Earth and Mars, focusing on habitability and conditions. Both planets share similarities, including polar caps, atmospheres, and water-related climates (Earth with oceans, Mars with ice). However, there are significant differences, with Mars being far less hospitable even compared to Earth's most extreme environments. In preparation for the Phoenix Mission, scientists conducted trial runs in Antarctica to simulate Martian conditions, studying trenching and ice-soil

boundaries. The lecture also covers various forms of water, or hydrology, found on the surface and underground on both planets, highlighting their contrasting environments.

本讲座将地球和火星进行比较，重点关注其宜居性和条件。两颗行星有相似之处，包括极地冰盖、大气层和与水有关的气候（地球有海洋，火星有冰）。然而，它们之间存在显著差异，与地球上最极端的环境相比，火星的适宜生存条件要差得多。在凤凰号任务的准备过程中，科学家在南极洲进行了试验，以模拟火星条件，研究挖沟和冰土边界。讲座还介绍了两颗行星表面和地下发现的各种形式的水或水文，突出了它们截然不同的环境。

32. Food Traffic Lights 食物标签 (萤火虫 28) (狸际 265)

Traffic light colours (red, amber and green) are used to represent food healthy standard. Different colours represent different information and categorize food types, so that people would know what to eat when they need some certain type of nutrients. Red food is high in things you should be cutting down on. Fine to have occasionally. Yellow food isn't high or low in bad nutrients, so an OK choice most of the time. Green food is the healthy choice, and the greener lights, the healthier the choice. It is the retailer's responsibility to label food properly so that consumers can choose exactly what type of food they need. In this way, consumers can be aware of food with less salt or less fat. This system can help the consumer to make decisions easier.

交通灯颜色（红色、琥珀色和绿色）用于表示食品健康标准。不同的颜色代表不同的信息，并将食物类型分类，这样人们在需要某种营养素时就知道该吃什么。红色食物富含你应该减少摄入的物质。偶尔吃一点没问题。黄色食物的不良营养成分含量不高也不低，所以大多数时候都是不错的选择。绿色食物是健康的选择，灯越绿，选择就越健康。零售商有责任正确标记食品，以便消费者可以准确选择他们需要的食品类型。这样，消费者就可以知道哪些食物含盐量少或脂肪量少。这个系统可以帮助消费者更容易做出决定。

33. Trade-offs Triangle 平衡三角 (多墨 1648) (萤火虫 71) (狸际 60)

The issue is achieving a win-win solution: promoting economic growth to reduce poverty without harming biodiversity. Protecting biodiversity as a primary goal risks affecting the poor and slowing economic growth. This debate is evident in developed, industrialized nations, where governments may seek to drill for oil, while conservationists argue against it for environmental reasons. The challenge lies in balancing economic development with ecological preservation to avoid negative impacts on both biodiversity and poverty reduction efforts.

问题在于找到一个双赢的解决方案：促进经济增长以减少贫困，同时不损害生物多样性。将保护生物多样性作为主要目标可能会影响穷人并减缓经济增长。这种争论在发达的工业化国家中很明显，这些国家的政府可能会寻求钻探石油，而环保主义者出于环境原因反对钻探石油。挑战在于平衡经济发展与生态保护，以避免对生物多样性和减贫努力产生负面影响。

34. Incentives for Salespeople 销售人员薪资机制 (多墨 1649) (萤火虫 25) (狸际 174)

Straight salary sales compensation plans, though uncommon, suit certain situations, like when direct sales are prohibited or when salespeople work in teams with equal contributions. This plan lacks incentives, often reducing motivation. Salary plus commission plans, the most common, offer a base salary plus commission, motivating productivity and providing stability. They work well when sales metrics ensure fair splits, though they are complex to administer. Commission-only plans pay solely based on sales achieved, attracting top performers but offering no income guarantee. While easy to administer, they can cause team aggression, high turnover, and burnout due to income insecurity.

直接工资销售薪酬计划虽然并不常见，但适用于某些情况，例如禁止直接销售或销售人员在团队中工作且贡献相同。这种计划缺乏激励，通常会降低积极性。最常见的是工资加佣金计划，提供基本工资加佣金，激励生产力并提供稳定性。当销售指标确保公平分配时，它们效果很好，尽管管理起来很复杂。仅佣金计划仅根据销售额支付报酬，吸引顶尖人才，但不提供收入保证。虽然易于管理，但它们可能导致团队侵略性、高流动率和收入不安全导致的倦怠。

35. Chest X-Ray 胸透视/生物医学工程 (多墨 1775) (萤火虫 33) (狸际 143)

This is one picture that probably you all know what it is when you see it. It's a familiar looking image. It's something that probably we all have some personal experience with, right? This is a chest x-ray that would be taken in your doctor's office or a radiologist's office. And it is a good example of biomedical engineering and that it takes a physical principle, that is how do x-rays interact with the tissues of your body, and it used that physics principle to develop a picture of what's inside your

body. And you will recognize some parts of the image, you can see ribcage, bones, the heart is the large bright object down here. If you have a good eyesight from the distance, you can even see the vessel leading out of the heart and into the lungs, and the lungs are darker spaces within the ribcage.

这张照片大家看到后可能都知道是什么。这是一张看起来很熟悉的图片。我们可能都有过亲身经历，对吧？这是一张胸部 X 光片，可以在医生办公室或放射科医生办公室拍摄。这是生物医学工程的一个很好的例子，它采用了物理原理，即 X 射线如何与身体组织相互作用，并利用该物理原理来生成身体内部的图片。你会认出图像的某些部分，你可以看到胸腔、骨头，心脏是下面那个大亮物体。如果你从远处视力好，你甚至可以看到从心脏通向肺部的血管，肺部为胸腔内较暗的空间。

36. Robots 机器人 (多墨 1776)

This lecture discusses the challenges of designing human-like robots. Robots struggle with tasks humans find simple because they lack sensory abilities, such as differentiating objects and colors. Unlike humans, robots cannot separate items by senses, making tasks like color recognition difficult. Artificial intelligence scientists have developed sensors with cameras to help robots identify colours by capturing pixels and transforming them into images. However, this process is complex, particularly in translating pixels accurately and distinguishing colours. Sensor improvements are necessary, highlighting the significant differences between humans and robots. There is no direct comparison between the two.

本讲座讨论了设计类人机器人的挑战。机器人很难完成人类认为简单的任务，因为它们缺乏感官能力，例如区分物体和颜色。与人类不同，机器人无法通过感官区分物品，这使得颜色识别等任务变得困难。人工智能科学家已经开发出带有摄像头的传感器，通过捕捉像素并将其转换为图像来帮助机器人识别颜色。然而，这个过程很复杂，特别是在准确转换像素和区分颜色方面。传感器的改进是必要的，这凸显了人类和机器人之间的显著差异。两者之间没有直接的可比性。

37. Edmund Wilson 威尔逊 (多墨 2068) (狸际 142)

Wilson emerged from a different background, becoming central to an American culture that valued accessible modern literature. Unlike abstract modernists, he believed literature should be appreciated by everyone, not just elites. Wilson played a key role in establishing innovative literature in American life, aiming to match Europe's great cultural achievements. With broad knowledge and high artistic standards, he believed authors like Proust, Joyce, Yeats, and Eliot should be accessible to ordinary Americans and worked to make that possible. Over nearly 50 years, Wilson was a literary journalist, investigative reporter, memoirist, and dedicated journal keeper.

威尔逊的背景与美国不同，他成为重视通俗易懂的现代文学的美国文化的核心人物。与抽象的现代主义者不同，他认为文学应该被所有人欣赏，而不仅仅是精英阶层。威尔逊在美国生活中建立创新文学方面发挥了关键作用，旨在与欧洲伟大的文化成就相媲美。凭借广博的知识和高超的艺术水准，他相信普鲁斯特、乔伊斯、叶芝和艾略特等作家应该让普通美国人能够读懂，并努力实现这一点。近 50 年来，威尔逊一直是一名文学记者、调查记者、回忆录作家和专职日记记录员。

38. Teaching 老教授谈教育 (多墨 2158) (萤火虫 26) (狸际 163)

All my research, conducted with over 60 graduate students, was driven by the desire to learn to teach. While some inventions occurred, I viewed them as byproducts of the learning process. The primary goal was always to gain deeper understanding or unify theory to improve teaching. I see teaching to test new ideas on engaged learners, making it a vehicle for sharing research results. This approach is stimulating and motivating for students, often revealing new research problems. The successful professor, in my view, unites research and teaching, benefiting from their close interplay.

我与 60 多名研究生一起进行的所有研究都是由学习以教学的愿望所驱动。虽然有些发明出现了，但我将它们视为学习过程的副产品。主要目标始终是获得更深入的理解或统一理论以改进教学。我认为教学是一种在积极学习者身上测试新想法的方式，使其成为分享研究成果的载体。这种方法对学生有激励作用，经常会揭示新的研究问题。在我看来，成功的教授将研究与教学结合起来，受益于两者的密切相互作用。

39. Happiness & Social Relation 幸福和社会关系 (萤火虫 142) (狸际 259)

Research suggests a positive correlation between happiness and the frequency and quality of social interactions. People tend to feel happier as their social connections increase and improve in quality. Satisfying social relationships contribute to greater happiness, which in turn encourages more social

engagement, especially with friends and family. However, it's uncertain whether social activities directly cause happiness or if naturally happy individuals are simply more inclined to be social. This relationship between happiness and social connection highlights the importance of quality interactions, though the exact causation remains unclear.

研究表明，幸福感与社交互动的频率和质量呈正相关。随着社交联系的增加和质量的提高，人们往往会感到更幸福。令人满意的社交关系有助于提高幸福感，进而鼓励人们更多地参与社交活动，尤其是与朋友和家人的社交活动。然而，尚不确定社交活动是否直接导致幸福感，或者天生快乐的人是否更倾向于社交。幸福感与社交联系之间的这种关系凸显了高质量互动的重要性，尽管确切的因果关系仍不清楚。

40. Civilization 文明 (猩际 235)

Our civilization is like a massive ship speeding into the future, more advanced and burdened than any before. While we can't predict every obstacle, understanding its direction, design, and crew abilities allows us to navigate wisely. Immediate action is crucial, as countless wrecks lie in our past, and this vessel is our last. Humanity's survival now depends on wise decisions in the coming years. Unlike other species, our immense impact means that mistakes are too costly. The world has become too small to tolerate large errors; our future hinges on responsible actions.

我们的文明就像一艘驶向未来的巨轮，比以往任何文明都更先进、负担更重。虽然我们无法预测每一个障碍，但了解它的方向、设计和船员的能力可以让我们明智地航行。立即采取行动至关重要，因为我们的过去曾发生过无数次沉船事故，而这艘船是我们的最后一艘。人类的生存现在取决于未来几年的明智决策。与其他物种不同，我们巨大的影响力意味着错误代价太高。世界已经变得太小，无法容忍大的错误；我们的未来取决于负责任的行动。

41. Library 图书馆 (多墨 3337)

You'll see the service desk after entering the library, and you can have these services at the service desk. There are computers on each floor, and you can use them to access the internet, check emails and library catalogs. Follow the orange signs to find printers and follow the instruction to use them. Remember to bring student cards, and you can charge the card to use the printer.

进入图书馆后会看到服务台，你可以在服务台享受这些服务。每层楼都有电脑，你可以用它们上网、查看电子邮件和图书馆目录。按照橙色标识找到打印机，按照指示使用。记得带上学生证，你可以用卡充值来使用打印机。

42. Bee Talk 蜜蜂交流 (多墨 3339)

Honeybees do a waggle dance to direct other bees to the sources of nectar. The dancing bees like this one can be halted by a headbutt from another bee. Now research has found that this headbutt is a warning signal. A feeding station was set up in the lab to mimic a source of nectar, then foraging bees were introduced to the dangers at the station, such as competition from rival colonies. When foragers returned to the hive, they stopped bees dancing. Scientists think the behavior warns dancers of dangerous sources of nectar.

蜜蜂会跳摇摆舞来引导其他蜜蜂找到花蜜来源。像这样的蜜蜂跳舞时，如果被另一只蜜蜂用头撞一下，就会停下来。现在研究发现，这种头撞是一种警告信号。在实验室中设立了一个喂食站来模拟花蜜来源，然后让觅食蜜蜂接触到喂食站的危险，例如来自竞争对手的竞争。当觅食蜜蜂返回蜂巢时，它们会停止跳舞。科学家认为，这种行为是在警告跳舞蜜蜂注意危险的花蜜来源。

43. Bilingual Education 双语教育 (猩际 311)

Many worry that raising children bilingually, especially in preschool years, may confuse them, as they believe the brain is naturally monolingual at birth. Parents often ask if using two languages at home will confuse their child. A common approach is for each parent to use a different language, helping the child associate each language with a specific person, thus reducing confusion. The concern is that if both parents switch between languages within conversations, the child might struggle to separate the languages. This fear drives parents to seek guidance on bilingual upbringing.

许多人担心，让孩子在双语环境中长大，尤其是在学龄前，可能会让他们感到困惑，因为他们认为大脑在出生时自然是单语的。父母经常问，在家里使用两种语言是否会让他们的孩子感到困惑。一种常见的方法是让父母各自使用不同的语言，帮助孩子将每种语言与特定的人

联系起来，从而减少困惑。令人担忧的是，如果父母双方在谈话中切换语言，孩子可能会难以区分这两种语言。这种恐惧促使父母寻求双语教育方面的指导。

44. Superman 超人 (多墨 3512) (萤火虫 44) (猩际 164)

Today, we explore tales of "hysterical strength," where people reportedly perform superhuman feats, like lifting cars off accident victims or moving heavy objects in emergencies. While many anecdotes support this phenomenon, we'll examine if adrenaline can truly grant such strength. Typically, these stories involve someone lifting a car, but factors like leverage and suspension springs may reduce the feat's difficulty. Most cases are anecdotal and often not repeatable, with people unable to replicate their strength later. However, there is substantial anecdotal evidence suggesting that, in moments of extreme crisis or fear, people may temporarily exhibit extraordinary strength.

今天，我们来探讨一下“歇斯底里的力量”的故事，据说人们会做出超人的壮举，比如从事故受害者身上抬起汽车或在紧急情况下移动重物。虽然许多轶事都支持这种现象，但我们将研究肾上腺素是否真的能赋予这种力量。通常，这些故事涉及有人举起汽车，但杠杆作用和悬架弹簧等因素可能会降低这一壮举的难度。大多数案例都是轶事，而且往往不可重复，人们以后无法复制他们的力量。然而，有大量轶事证据表明，在极度危机或恐惧的时刻，人们可能会暂时表现出非凡的力量。

45. Clouds Formation 云的形成 (多墨 3544) (萤火虫 54) (猩际 249)

Clouds are formed around particles, which serve as nuclei for water condensation. Sea spray and sea salt are effective nuclei, especially over oceans, while pollution influences cloud formation inland. Different sources affect cloud formation rates, and scientists are investigating which contribute most. Clouds are crucial in climate regulation, as they reflect sunlight back into space, helping to cool the Earth. They also impact regional weather; increased pollution in clouds can alter weather patterns and reduce precipitation, leading to drought in areas with high air pollution. Understanding cloud formation is key to studying climate and weather changes.

云是在粒子周围形成的，粒子是水凝结的核心。海浪和海盐是有效的核心，尤其是在海洋上空，而污染则影响内陆的云层形成。不同的来源影响云的形成率，科学家正在研究哪些因素影响最大。云在气候调节中至关重要，因为它们将阳光反射回太空，有助于冷却地球。它们还影响区域天气；云中污染的增加会改变天气模式并减少降水，导致空气污染严重的地区出现干旱。了解云的形成是研究气候和天气变化的关键。

46. Aristotle on Rhetoric and Truth 真理和修辞 (萤火虫 132) (猩际 258)

Aristotle argues that rhetoric is necessary to influence people, particularly those unable to grasp truth directly. He distinguishes truth from rhetoric, with truth being abstract and without form, while rhetoric is its "body" or presentation. To lead others to truth, one may need to use rhetorical "tricks" because most people, in Aristotle's view, lack the ability to perceive truth on their own. Only the educated can truly understand it. Rhetoric, therefore, serves as a tool to guide the general public toward understanding, acting as a form of persuasive logic.

亚里士多德认为，修辞对于影响人们，特别是那些无法直接掌握真理的人是必不可少的。他将真理与修辞区分开来，真理是抽象的、没有形式的，而修辞是真理的“主体”或表现形式。要引导他人走向真理，可能需要使用修辞“技巧”，因为在亚里士多德看来，大多数人缺乏自己感知真理的能力。只有受过教育的人才能真正理解它。因此，修辞是一种引导公众走向理解的工具，是一种有说服力的逻辑。

47. Rice 大米 (多墨 3602)

The Green Revolution began in 1943 when Mexico sought help to feed its population. The Ford and Rockefeller Foundations established the International Rice Research Institute, which led to the development of IR8 rice by 1962. IR8 was a modified crop created through selective breeding, not genetic engineering, and produced five times the yield of traditional rice, or ten times with fertilizer. By 1980, IR36 further improved yields by allowing two harvests per year, and by 1990, IR72 surpassed IR36 using advanced genetic techniques. From 1960 to 2000, the Green Revolution significantly boosted global crop yields.

绿色革命始于 1943 年，当时墨西哥寻求帮助来养活其人口。福特和洛克菲勒基金会成立了国际水稻研究所，并于 1962 年开发出 IR8 水稻。IR8 是一种通过选择性育种而非基因工程创造的改良作物，产量是传统水稻的五倍，使用化肥后产量是传统水稻的十倍。到 1980

年，IR36 通过允许每年两次收获进一步提高了产量，到 1990 年，IR72 使用先进的基因技术超越了 IR36。从 1960 年到 2000 年，绿色革命显著提高了全球农作物产量。

48. Venus 金星 (多墨 3630) (萤火虫 80) (猩际 286)

Before the space age, artists imagined Venus as a lush, Earth-like planet, blending science fiction and fact. This view persisted until 1967, when the first detailed measurements of Venus were made. Mariner 4 and 5 revealed that Venus's surface was extremely hot, with clouds of sulfuric acid, disproving the idea of a water cycle. These in-situ spacecraft measurements showed that Venus was far less Earth-like than once thought. While Venus has some Earth-like features, as confirmed by Venus Express, it remains vastly different from Earth overall.

在太空时代之前，艺术家们将金星想象成一个郁郁葱葱的类地行星，将科幻与现实融为一体。这种观点一直持续到 1967 年，当时首次对金星进行了详细测量。水手 4 号和 5 号发现金星表面温度极高，有硫酸云，推翻了水循环的理论。这些航天器现场测量表明，金星与地球的相似程度远低于人们曾经认为的。虽然金星具有一些与地球类似的特征，正如金星快车号所证实的那样，但它与地球的整体差异仍然很大。

49. Practice and Performance 熟能生巧 (萤火虫 61) (猩际 237)

Deliberate practice is essential for achieving high performance in any field. It requires time, effort, and the ability to identify and address weaknesses. For example, to become a skilled violinist, one must dedicate thousands of hours to solitary practice. Similarly, in subjects like geometry, focusing on areas of difficulty can improve understanding. Even those with natural talent often need years of deliberate practice to reach international recognition. This commitment to consistent improvement highlights the value of practice, reminding us that it is a critical step toward success and should never be a source of shame.

刻意练习对于在任何领域取得优异成绩都至关重要。它需要时间、精力以及发现和解决弱点的能力。例如，要成为一名熟练的小提琴手，必须投入数千小时进行独自练习。同样，在几何等学科中，专注于困难领域可以提高理解力。即使是天赋异禀的人也常常需要多年的刻意练习才能获得国际认可。这种对持续改进的承诺凸显了练习的价值，提醒我们练习是迈向成功的关键一步，永远不应成为耻辱的根源。

50. Linguists 语言学家 (多墨 3730) (猩际 247)

Linguists often gain "invisible training" that positions them as authorities on language. However, when working with communities, it is essential to recognize that the true authority over language resides with the community itself. Expertise, gained through formal training, differs from authority, which is inherently tied to cultural and community ownership. A student in Sydney highlighted this distinction, emphasizing that while linguists have specialized knowledge, they do not hold the right to dictate language use. Language loss often results from cultural dominance, and real language revival requires restoring authority to communities, not just applying external expertise.

语言学家通常会接受“隐形培训”，从而成为语言方面的权威。然而，在与社区合作时，必须认识到语言的真正权威在于社区本身。通过正规培训获得的专业知识不同于权威，权威本质上与文化和社区所有权有关。悉尼的一名学生强调了这一区别，强调虽然语言学家拥有专业知识，但他们无权决定语言的使用。语言的消失往往是文化主导的结果，真正的语言复兴需要恢复社区的权威，而不仅仅是应用外部专业知识。

51. Government Blogging 政府博客 (萤火虫 43) (猩际 245)

We normally see blogging as a two-way interaction, in which the blogger creates the content, and the readers interact or challenge the author. But the case will be much difficult when it comes to government, such as the white house. Because people will become coarser and ruder online, especially in the comment area. Hence the governor blog may go wild and chaotic. So, the government will have to administrate the comment. Once the government starts administrating the comment, citizens may find the government manipulating what should be said and what should be shown, which contradicts the original intention.

我们通常认为博客是一种双向互动，博主创作内容，读者与作者互动或挑战作者。但是，当涉及到政府，比如白宫时，情况会变得困难得多。因为人们会在网上变得更加粗鲁和无礼，尤其是在评论区。因此，州长博客可能会变得狂野和混乱。因此，政府将不得不管理评论。一旦政府开始管理评论，公民可能会发现政府操纵了应该说什么和应该展示什么，这与初衷相悖。

52. A Survey 一项调查 (多墨 4086) (猩际 248)

Let's say if I'm asking which source, you often use to get information. Newspaper? Radio? TV? And the survey shows 62% of the people chose the internet. You might be thinking I am going to say, how important the internet is, or how quickly it has changed the world for a few years. But what if I tell you this survey is conducted on the website global and mail. Our answer will be different. Because the people who did this survey on a website must be frequent users of the internet. This sample is a biased sample. So, we must pay attention to how a survey is conducted.

假设我问你经常使用哪种来源获取信息。报纸？广播？电视？调查显示，62% 的人选择了互联网。你可能会想我会说，互联网有多重要，或者它在几年内改变了世界有多快。但如果我告诉你这项调查是在 global and mail 网站上进行的。我们的答案会有所不同。因为在网站上进行这项调查的人一定是互联网的频繁用户。这个样本是有偏差的样本。所以，我们必须注意调查是如何进行的。

53. Soot Emission / Air Pollution 尘排放/空气污染 (萤火虫 10) (猩际 223)

Small things can have big impacts, like soot, also known as black carbon, which is released from burning fuel, coal, and wood. Soot causes respiratory issues and 1.6 million premature deaths annually, mainly affecting the poor. It travels far, reaching places like the Himalayas and Arctic, where it absorbs sunlight, accelerating snowmelt and raising sea levels, threatening wildlife and communities. Soot is the second-largest contributor to Arctic warming after CO₂, causing 25% of human-induced warming. Reducing black carbon could quickly slow global warming, buying time for the Arctic. Cleaner technologies and international action can make a big difference.

小事也能造成大影响，比如烟尘，又称黑碳，是燃料、煤炭和木材燃烧时释放的。烟尘每年导致呼吸问题和 160 万人过早死亡，主要影响穷人。烟尘传播范围很广，到达喜马拉雅山和北极等地，在那里吸收阳光，加速融雪并抬高海平面，威胁野生动物和社区。烟尘是继二氧化碳之后导致北极变暖的第二大因素，占人类活动导致的变暖的 25%。减少黑碳可以迅速减缓全球变暖，为北极争取时间。更清洁的技术和国际行动可以发挥巨大作用。

54. Happiness 关于幸福的政策 (多墨 4268)

Only Bhutan has adopted Gross National Happiness (GNH) as its central policy index, achieving success in education, health, economic growth, and environmental preservation. Bhutan uses a sophisticated system to measure how policies impact happiness. Now, other countries like Australia, France, and Great Britain are showing interest in happiness research, exploring its potential influence on public policy. These countries are considering publishing regular happiness statistics, indicating growing interest among policymakers and legislators. The idea of integrating happiness into national policy is gradually gaining attention in advanced nations.

只有不丹将国民幸福总值 (GNH) 作为其核心政策指标，并在教育、健康、经济增长和环境保护方面取得了成功。不丹使用一套复杂的系统来衡量政策如何影响幸福感。现在，澳大利亚、法国和英国等其他国家也对幸福感研究表现出兴趣，探索其对公共政策的潜在影响。这些国家正在考虑定期发布幸福感统计数据，这表明政策制定者和立法者对此的兴趣日益浓厚。将幸福感纳入国家政策的想法正在发达国家逐渐引起关注。

55. Melatonin 褪黑素 (猩际 188)

I'm just going to take on the stuff were left off. The hormone I want to talk about is called melatonin. The synthesis is in the pineal gland, which is very small. It is the size of a pea in your brain. The corpus is the site of the soul, and it is where melatonin is made. And it has a rhythm as well. And in the sense, it is the opposite of the cortisol. It peaks at night. We call it the darkness hormone. In every species that we studied, melatonin occurs at night. And its hormone that prepares you for the things, that your species, does at night. So, of course, in humans, we sleep, but animals, like a rodent, they are awake. It's a hormone that is related to darkness behavior.

我现在要继续讲讲刚才讲过的内容。我现在要讲的荷尔蒙叫做褪黑激素。褪黑激素是在松果体中合成的，松果体很小，大小和你大脑里的豌豆差不多。脑体是灵魂的所在，也是褪黑激素的产生地。褪黑激素也有节律。从这个意义上说，它与皮质醇相反。它在夜间达到峰值。我们称之为黑暗激素。在我们研究的每个物种中，褪黑激素都在夜间产生。褪黑激素是一种让你为你的物种在夜间所做的事情做好准备的激素。所以，当然，人类是睡觉的，但动物，比如啮齿动物，它们是清醒的。褪黑激素是一种与黑暗行为有关的激素。

56. Science and Scientist 科学与科学家 (猩际 261)

The author plans to write about both science and scientists in his book, though he finds scientists more intriguing. He compares science to complex furniture, built on evidence, while scientists bring stories, conduct research, and seek truth, often finding themselves in fascinating situations. Despite the scientific themes, he believes the book will be accessible to general readers, as it's a science fiction work rather than a strict scientific text. Readers don't need advanced knowledge to enjoy it. Unlike scientific journals, his fiction will include relatable narratives, such as stories of "happy families," making it engaging for a broader audience.

作者计划在书中同时写科学和科学家，但他觉得科学家更有趣。他将科学比作建立在证据之上的复杂家具，而科学家则带来故事、进行研究并寻求真相，他们常常发现自己身处令人着迷的境地。尽管主题是科学，但他相信这本书对普通读者来说也是可以理解的，因为它是一部科幻作品，而不是严格的科学文本。读者不需要高级知识就可以欣赏它。与科学期刊不同，他的小说将包括可理解的叙述，例如“幸福家庭”的故事，从而吸引更广泛的读者。

57. Caterpillar / Early Spring 早春 (多墨 4444)

The temperature is rising, making springtime phenomena earlier. Urban light pollution is also a more significant factor. This disruptive timing can damage the ecosystem. Caterpillars feed on the tree, want to feed on the juiciest and least chemically protected leaves. Nesting birds hatch their eggs at the same time for the maximum number of caterpillars. This could affect the survival of birds. This disruptive timing can ripple through the ecosystem. But smarter lighting—like LEDs that dial down certain wavelengths—could help.

气温不断升高，春天的现象越来越早。城市光污染也是一个更重要的因素。这种破坏性的时间安排会破坏生态系统。毛毛虫以树为食，它们想吃最鲜美、化学保护最少的叶子。筑巢的鸟儿同时孵蛋，以产出最多的毛毛虫。这可能会影响鸟类的生存。这种破坏性的时间安排会对生态系统产生连锁反应。但更智能的照明——比如调低某些波长的 LED——可能会有所帮助。

58. Hazardous Chemical Product Label 危险化学品警示标签 (多墨 4477) (萤火虫 56)

The design of hazard chemical labels is concerning. Often, these labels display attractive images, such as grassy fields, which can mislead consumers and downplay the product's dangers. The warning information is typically small and only on the back, making it hard to read. Studies show that 60% of Americans struggle to recognize fonts smaller than size 10, and 40% are illiterate. This lack of clarity means many people are unaware of the hazards posed by these chemicals, increasing the risk of misuse. Clear, prominent warnings are essential to ensure consumer safety and prevent dangerous misunderstandings.

危险化学品标签的设计令人担忧。这些标签上通常显示有吸引力的图像，例如草地，这可能会误导消费者并淡化产品的危险性。警告信息通常很小，并且只在背面，难以阅读。研究表明，60% 的美国人难以识别小于 10 号的字体，40% 的人是文盲。这种缺乏清晰度意味着许多人不知道这些化学品带来的危害，增加了误用的风险。清晰、突出的警告对于确保消费者安全和防止危险的误解至关重要。

59. Animal Behaviour 动物行为 (多墨 4486)

Animal behaviour can be studied through two fundamental types of questions: proximate and ultimate. Proximate questions focus on mechanisms behind behaviour, asking how it develops and what causes it. Ultimate questions explore the evolution of behaviour, examining how it evolved and its adaptive significance. These four questions, known as Tinbergen's four questions—developed by Niko Tinbergen, a pioneer in animal behaviour—offer a comprehensive framework for studying behaviour. Understanding the differences between these questions is essential for studying animal behaviour and biology, with the approach depending on the specific question being addressed.

动物行为可以通过两种基本问题来研究：近因问题和终极问题。近因问题关注行为背后的机制，询问行为如何发展以及原因是什么。终极问题探索行为的进化，研究行为如何进化及其适应性意义。这四个问题被称为丁伯根四个问题——由动物行为学先驱尼科·丁伯根提出——为研究行为提供了一个全面的框架。了解这些问题之间的差异对于研究动物行为和生物学至关重要，方法取决于要解决的具体问题。

60. Smell of Books 书的味道 (多墨 4555)

Historic libraries have a unique, recognizable smell that holds significance for visitors, creating a deep connection to the environment. At Saint Paul's Cathedral's Wren Library, the UCL Centre for

Sustainable Heritage was asked to not only assess the environment but also preserve the library's distinct smell, as it is important to the visitor experience. Research indicates that smell plays a crucial role in how we interact with our surroundings, second only to sight. This insight, valued by advertisers, highlights the importance of scent in shaping our perception and enhancing cultural experiences.

历史悠久的图书馆具有独特的、可识别的气味，对访客来说意义重大，与环境建立了深厚的联系。在圣保罗大教堂的雷恩图书馆，伦敦大学学院可持续遗产中心不仅被要求评估环境，而且还要保留图书馆独特的气味，因为这对访客体验很重要。研究表明，嗅觉在我们与周围环境的互动中起着至关重要的作用，仅次于视觉。广告商重视这一见解，强调了气味在塑造我们的感知和增强文化体验方面的重要性。

61. Shy Fish 腼腆的鱼/胆小鱼实验 (多墨 4556) (萤火虫 126) (猩际 153)

A study on stickleback fish reveals that shy individuals prefer to follow similarly timid leaders, while bold fish are indifferent to the leader's personality. In experiments, trios of sticklebacks with known personalities played "follow the leader" in a tank with food hidden at one end. Shy fish emerged more often when a fellow shy fish led, while bold followers showed no preference. However, bold fish led more frequently overall, initiating more trips regardless of followers. This suggests that sticklebacks prefer leaders matching their personality, though bold individuals often lead through sheer persistence. These findings are published in *Biology Letters*.

一项关于棘鱼的研究表明，害羞的个体更喜欢跟随同样胆小的领导者，而大胆的鱼则对领导者的性格漠不关心。在实验中，三条性格已知的棘鱼在一个水箱里玩“跟随领导者”的游戏，水箱的一端藏有食物。当有一条害羞的鱼带领时，害羞的鱼会更频繁地出现，而大胆的追随者则没有表现出任何偏好。然而，大胆的鱼总体上更频繁地带领，无论有没有人跟随，它们都会发起更多的旅行。这表明棘鱼更喜欢与它们性格相符的领导者，尽管大胆的个体往往凭借纯粹的坚持来领导。这些发现发表在《生物学快报》上。

62. Cracking Sound of Knee 膝盖的声音 (多墨 4568) (萤火虫 125) (猩际 272)

The unpleasant sound of a cracking knee may hold valuable information for rehabilitation, says Omer Inan, an electrical engineer at Georgia Tech. Inan's interest began in college as an athlete, where he noticed knee creaking after intense workouts. Later, inspired by DARPA's call for better knee injury tech, he and his team developed a prototype using tiny microphones and sensors on stretchy tape to record knee sounds. Tests suggest injured knees produce more erratic sounds than healthy ones, which could help patients track healing post-surgery. This research, published in *IEEE Transactions on Biomedical Engineering*, aims to assist rehab decisions.

佐治亚理工学院的电气工程师 Omer Inan 表示，膝盖开裂时发出的难听声音可能为康复提供宝贵信息。Inan 的兴趣始于大学时，当时他是一名运动员，在剧烈运动后，他注意到膝盖会发出吱吱声。后来，受到 DARPA 呼吁改进膝盖损伤技术的启发，他和他的团队开发了一个原型，使用弹性胶带上的微型麦克风和传感器来记录膝盖声音。测试表明，受伤的膝盖比健康的膝盖发出更多不规则的声音，这可以帮助患者跟踪术后康复情况。这项研究发表在《IEEE 生物医学工程学报》上，旨在协助康复决策。

63. Sugar 糖 (多墨 4994) (猩际 148)

Sugar is present in many foods where you might not expect it, beyond sweets like donuts, ice cream, and candy. For example, Skippy Peanut Butter lists sugar as its second ingredient, right after peanuts, indicating a high amount. Similarly, even foods like beef stew contain sugar. Although it appears lower on the ingredient list, there's actually more sugar than carrots, despite the label highlighting fresh vegetables. This hidden sugar content in unexpected items like peanut butter and beef stew shows how common added sugars are in various foods.

除了甜甜圈、冰淇淋和糖果等甜食之外，糖还存在于许多你意想不到的食物中。例如，Skippy 花生酱将糖列为第二种成分，仅次于花生，表明糖的含量很高。同样，即使是炖牛肉这样的食物也含有糖。虽然它在配料表中的位置较低，但实际上它的含糖量比胡萝卜还高，尽管标签上强调的是新鲜蔬菜。花生酱和炖牛肉等意想不到的食物中隐藏的糖含量表明，各种食物中添加糖的现象非常普遍。

64. Music and Well-being 音乐和幸福 (多墨 5016) (猩际 152)

Well-being encompasses health, happiness, achievement, and a thriving state of mind and body. It applies to everyone, not just those facing challenges. Music plays an intuitive role in managing well-being—lifting spirits, calming, and healing. Our research aims to move from this intuitive use of

music to an informed application in communities, enhancing our understanding of music's impact on human life. Music already benefits us by soothing infants, uniting communities, and inspiring work and personal growth. But we believe there's even more potential to explore in music's contribution to well-being.

幸福感包括健康、快乐、成就以及身心健康。幸福感适用于所有人，而不仅仅是那些面临挑战的人。音乐在管理幸福感方面发挥着直观的作用——振奋精神、镇静和治愈。我们的研究旨在从这种直观的音乐使用转向在社区中的明智应用，增强我们对音乐对人类生活的影响的理解。音乐已经通过安抚婴儿、团结社区、激励工作和个人成长等方式使我们受益。但我们相信，音乐对幸福感的贡献还有更大的潜力值得探索。

65. Early Robots 早期机器人 (多墨 5025) (萤火虫 46) (猩际 124)

The concept of robots originated in the 1920s play *Rossum's Universal Robots* by Czech writer Karel Capek. Unlike today's view of robots as cute or apolitical, these early robots were created in a context of political upheaval following World War I. The play explores questions of humanity and identity. In it, robots are human-like but are assembled on a production line, like Ford's manufacturing process. These robots were designed primarily for labor, reflecting societal concerns about human roles and mechanization during that turbulent period.

机器人的概念起源于捷克作家卡雷尔·恰佩克 20 世纪 20 年代的戏剧《罗森的万能机器人》。与今天对机器人可爱或不关心政治的看法不同，这些早期的机器人是在第一次世界大战后的政治动荡背景下创造的。该剧探讨了人性和身份问题。在剧中，机器人与人类相似，但在生产线上组装，类似于福特的制造过程。这些机器人主要用于劳动，反映了社会对那个动荡时期人类角色和机械化的担忧。

66. Pavlov Experiment 巴普洛夫实验 V2 (多墨 5439) (猩际 165)

In this discussion, I'll explore the emotions we share with animals and new brain imaging technology, like functional magnetic resonance imaging (fMRI), to address old questions about motivation and emotion. Let's consider Pavlov's classic experiment from over a century ago, where a dog was conditioned to associate a sound with food, leading to increased salivation and movement. Beyond physical responses, Pavlov's work hints at the brain's role in creating a "competitive state." This state reflects both physiological reactions and emotional anticipation, much like how we might feel before a meal today.

在本次讨论中，我将探讨我们与动物共有的情感以及新的大脑成像技术，如功能性磁共振成像 (fMRI)，以解决有关动机和情感的旧问题。让我们考虑一下巴甫洛夫一个多世纪前的经典实验，其中一条狗被训练将声音与食物联系起来，导致唾液分泌增加和运动增加。除了身体反应之外，巴甫洛夫的工作还暗示了大脑在创造“竞争状态”方面的作用。这种状态既反映了生理反应，也反映了情绪预期，就像我们今天饭前的感觉一样。

67. King 国王 (多墨 5931) (猩际 281)

At the top of the feudal system was the king, who ruled a kingdom. Governing was challenging, especially in the Middle Ages, and the king often owed favors to those who helped him gain power. To manage his realm, he granted land (the key currency) in exchange for loyalty and service. A king might give a duchy to a duke, who would pledge loyalty and provide troops in times of war or for conquests. The duke also paid taxes to the king, either in coin or agricultural produce, depending on the period. This system helped the king maintain control over his kingdom.

封建制度的最高层是国王，统治着一个王国。统治是一个挑战，尤其是在中世纪，国王经常欠那些帮助他获得权力的人恩惠。为了管理他的王国，他授予土地（主要货币）以换取忠诚和服务。国王可能会将公国赠给公爵，公爵会宣誓效忠并在战争或征服时期提供军队。公爵还向国王纳税，根据时期的不同，可能是硬币，也可能是农产品。这种制度帮助国王保持对王国的控制。

68. Globalization 全球化 (猩际 276)

Reflecting on globalization, it's evident how the world has changed over the past few decades. Previously, events like a chicken dying of illness in East Asia or a bank failure in North America would have been local tragedies, not global threats. Now, such incidents can trigger pandemics or economic crises worldwide. Globalization allows us to share ideas, knowledge, and culture more rapidly and affordably, bringing both challenges and benefits. Notably, the Millennium Development

Goals demonstrated humanity's potential for progress, achieving several targets early, showing that when united, we can accomplish remarkable feats.

回顾全球化，不难发现过去几十年来世界发生了怎样的变化。以前，东亚一只鸡因病死亡或北美一家银行倒闭等事件只是地方悲剧，而不是全球威胁。现在，这些事件可能引发全球流行病或经济危机。全球化使我们能够更迅速、更经济地分享思想、知识和文化，既带来挑战，也带来好处。值得注意的是，千年发展目标展示了人类进步的潜力，提前实现了几个目标，表明只要团结起来，我们就能取得非凡成就。

69. A Successful Entrepreneur 成功企业家 (多墨 7641)

To succeed as an entrepreneur, having good ideas is crucial, but a "good idea" has specific qualities. A great idea should be novel, unique, and transformative. Although ideas often combine smaller concepts, they can still be distinctive if infused with individual creativity rather than mere copying. Constant innovation prevents stagnation, enhancing the idea's uniqueness. Unique concepts often extend basic notions, surpassing predecessors by serving their purpose more effectively. Thus, originality and continuous improvement are key to making an idea impactful and successful.

要想成为一名成功的企业家，拥有好的想法至关重要，但“好想法”具有特定的特质。一个伟大的想法应该是新颖、独特和具有变革性的。尽管想法通常结合了较小的概念，但如果注入个人创造力而不是单纯的复制，它们仍然可以与众不同。不断创新可以防止停滞不前，增强想法的独特性。独特的概念通常会扩展基本概念，通过更有效地实现其目的而超越前人。因此，原创性和持续改进是让一个想法具有影响力和成功的关键。

70. Red Planet 火星有水? (萤火虫 21) (猩际 73)

Mars, the fourth planet from the Sun and second smallest after Mercury, is called the "Red Planet" due to iron oxide on its surface. With a thin atmosphere, it has craters, volcanoes, valleys, deserts, and polar ice caps. Mars has rotational periods and seasonal cycles similar to Earth's. Research indicates Mars once had liquid water, supported by findings of calcium carbonate, salt, minerals, and perchlorate—elements essential for water formation. This evidence suggests that Mars may have been hospitable long ago, with significant surface water in its early history.

火星是距太阳第四远的行星，也是仅次于水星的第二小行星，由于表面含有氧化铁而被称为“红色星球”。火星大气层稀薄，有陨石坑、火山、山谷、沙漠和极地冰盖。火星的自转周期和季节循环与地球相似。研究表明，火星曾经有液态水，碳酸钙、盐、矿物质和高氯酸盐等水形成所必需的元素也证实了这一点。这些证据表明，火星可能很久以前就适宜居住，在其早期历史中曾有大量地表水。

71. Four Fundamental Forces 四种基本力 (多墨 8555)

The universe operates through four fundamental forces. Gravity, which keeps us grounded and holds planets in orbit, is one. Electromagnetic force, responsible for electricity, light, and solar radiation, is another familiar force. The less-known nuclear forces work within atomic nuclei. The strong nuclear force binds protons and quarks together, preventing them from separating, while the weak nuclear force causes radioactive decay. Together, these four forces—gravity, electromagnetic, strong nuclear, and weak nuclear—govern interactions in the universe, shaping everything from atomic behaviour to planetary motion.

宇宙通过四种基本力运转。引力是其中一种，它让我们脚踏实地，让行星在轨道上运行。电磁力是另一种常见的力，它产生电、光和太阳辐射。鲜为人知的核力在原子核内起作用。强核力将质子和夸克结合在一起，防止它们分离，而弱核力则导致放射性衰变。这四种力——引力、电磁力、强核力和弱核力——共同控制着宇宙中的相互作用，塑造了从原子行为到行星运动的一切。

72. Universal Philosophy 普遍哲学 (多墨 8667)

A universal philosophy claims to apply to everyone, across all countries, which may seem unusual. However, liberal theory argues that certain value structures are indeed universal, as seen in institutions like the United Nations and the Universal Declaration of Human Rights, based on shared human values. This concept is also reflected in sports: while football has core rules everyone follows, different countries have unique styles. Brazilian, Italian, and British football each show cultural distinctions, yet adhere to universal rules. Thus, universal values coexist with cultural variations within a shared framework.

一种普遍的哲学声称适用于所有国家/地区的每个人，这似乎有些不寻常。然而，自由主义理论认为，某些价值结构确实是普遍的，正如联合国和《世界人权宣言》等机构所见，它们基于人类共同的价值观。这一概念也反映在体育运动中：虽然足球有每个人都遵循的核心规则，但不同国家有独特的风格。巴西、意大利和英国足球各有文化差异，但都遵循普遍规则。因此，普世价值观与文化差异在共同框架内共存。

73. Licking and Grooming 舔舐和梳理毛发 (萤火虫 6) (猩际 75)

The study examines how maternal licking and grooming (LG) affects adult rats' stress reactivity. LG reflects the care mother rats provide to their offspring. Findings show that high LG leads to moderate stress reactivity in adult rats, suggesting that a nurturing environment in infancy can reduce risks of developmental issues and diseases later. Conversely, low LG increases stress reactivity, raising risks for conditions like heart disease, diabetes, and affective disorders. This research highlights the critical role of maternal care in shaping stress response systems, emphasizing the importance of supportive environments for healthy development and long-term health.

该研究调查了母鼠舔舐和梳理 (LG) 如何影响成年大鼠的压力反应。LG 反映了母鼠对后代的关爱。研究结果表明，较高的 LG 会导致成年大鼠出现中等程度的压力反应，这表明婴儿时期的养育环境可以降低成年大鼠出现发育问题和疾病的风险。相反，较低的 LG 会增加压力反应，增加患心脏病、糖尿病和情感障碍等疾病的风险。这项研究强调了母鼠护理在塑造压力反应系统中的关键作用，强调了支持性环境对健康发展和长期健康的重要性。

74. Solar Energy 太阳能 (萤火虫 111) (猩际 325)

The lecture discusses solar energy, highlighting its benefits and challenges. The main advantage is its limitless supply, alleviating concerns about resource depletion. However, solar energy has drawbacks, including high costs and limited availability since it cannot be used 24/7. To address this, scientists suggest storing solar energy by converting it into chemical fuel, enabling energy use even when sunlight is unavailable. Research is ongoing to find methods for utilizing solar energy at night, further enhancing its practicality and efficiency as a sustainable energy source.

讲座讨论了太阳能，强调了其优势和挑战。太阳能的主要优势在于其供应无限，缓解了人们对资源枯竭的担忧。然而，太阳能也有缺点，包括成本高、可用性有限，因为它不能全天候使用。为了解决这个问题，科学家建议将太阳能转化为化学燃料来储存，这样即使在没有阳光的情况下也能使用能源。目前正在进行研究，以寻找在夜间利用太阳能的方法，进一步提高其作为可持续能源的实用性和效率。

75. Electric Vehicles 电动汽车 (猩际 9)

Electric vehicles (EVs) were among the earliest motorized vehicles and were more popular than petrol cars in the early 20th century due to their comfort. However, as petrol-powered cars advanced, EVs became limited to specific uses like forklifts and ambulances. While electricity use in cars declined, it became essential for trains, with Switzerland leading in electrified train systems due to a lack of coal or oil. Today, interest in EVs has resurged due to rising oil costs and environmental concerns. Car emissions contribute to ozone depletion and polar warming, prompting research into effective electric transport solutions.

电动汽车 (EV) 实际上是最早的机动车之一，由于其舒适性，在 20 世纪初比汽油车更受欢迎。然而，随着汽油动力汽车的发展，电动汽车仅限于叉车和救护车等特定用途。虽然汽车用电量下降，但电力对火车来说却必不可少，瑞士由于缺乏煤炭或石油，在电气化火车系统方面处于领先地位。如今，由于石油成本上升和环境问题，人们对电动汽车的兴趣再次高涨。汽车排放导致臭氧层消耗和极地变暖，促使人们研究有效的电动运输解决方案。

76. Software 软件 (猩际 326)

The history of software is quite recent, with the IT industry only around 67 years old. The rapid advancement, driven by Moore's Law, has led to an explosion in computing power and a drastic reduction in costs, allowing IT to penetrate areas of life previously unimaginable. This progress resembles a "stampede" as people have focused on seizing "low-hanging fruit" in software development. This rush to exploit immediate opportunities has shaped the evolution of software and continues to influence its direction, bringing various consequences that warrant further exploration.

软件的历史相当短暂，IT 行业只有 67 年左右的历史。在摩尔定律的推动下，软件的快速发展带来了计算能力的爆炸式增长和成本的大幅降低，使 IT 能够渗透到以前无法想象的生活中的各个领域。这种进步就像一场“踩踏”，因为人们专注于在软件开发中抓住“唾手可得的果实”。这种抢占眼前机会的冲动塑造了软件的发展，并继续影响其方向，带来了各种值得进一步探索的后果。